

PAPER_2130_UNIFIED_REGISTRY_PROGRAM_COMPLETE_R0_R5_ONE_REGISTRY_ONE_CORPUS_PASS_ALL_CONSTANTS_2544_ROWS_656_EDGE_FALSIFIABILITY_GRAPH_THREE_LANGUAGE_PINS_UQFF_LANDMARK

Daniel T. Murphy

PAPER_2130 — The Unified Registry Program Complete (R0–R5): One Registry, One Corpus Pass, All Constants

Author: Daniel T. Murphy

Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.77+

Date: 2026-07-24

Landmark Type: Program Completion — corpus-scale curation, canonicalization, and falsifiability infrastructure

Program: UNIFIED_REGISTRY_PROGRAM (UNIFIED_REGISTRY_PROGRAM_PLAN.md), phases R0–R5, executed 2026-07-22 → 2026-07-24

Status: Formal landmark whitepaper — UQFF canonical

Abstract

The Unified Registry Program converted the Star-Magic corpus — 2,227+ whitepapers, 1,321 dispatch closures, 1,392 over-determination observables, multi-language implementations (Python, C++, Lean 4) — from a large, redundant archive into a **curated replication ledger** with a single source of truth. Six phases: R0 built the 2,544-row UNIFIED_REGISTRY.csv by joining every existing ledger family (all seven protected baselines preserved bit-identical, hash-pinned); R1 adjudicated canonical routes (109 explicit verdicts + 2,435 sole-route auto-canonicalizations, 4 rulings by the author); R2 executed one append-only corpus pass (199 registry-keyed derivation notes; pdf2 brought to full 2,226-PDF parity); R3 collapsed all per-class constant definitions into uqff_registry_primitives.py with three-language agreement pins (Python = C++ = Lean); R4 computed the 656-edge falsifiability graph answering "*if primitive X were revised, what breaks*"; R5 made the registry the production status-report backend and preprint results table. The framework now states, with machine-checked backing: **9 independent primitives, 14 live-composed derived constants (7 EXACT), 2,544 registered quantities, 656 falsifiability edges, and a fidelity gate above 3,180 assertions — regenerable end-to-end by four idempotent scripts.**

1. Why the program existed

G and c were not special. The corpus contained derivations for essentially every constant (the 1209-series AA–HH alone covers k_B, h, c, e, N_A, F, σ , b_Wien, E_R, E_h), 1,000+ paradox/dispatch closures, multiple independent routes to the same quantities, and triple-language implementations. Per-constant campaigns would have re-scanned 2,227 markdowns N times. The program's unit of work was:

> **One registry. One canonicalization pass. One corpus pass. One code source-of-truth. One dependency graph.**

The redundancy was never waste. Each re-derivation is an independent replication; each multi-route constant is over-determination; each cross-language agreement is verification. The program **curated the redundancy into a replication ledger** rather than removing it.

2. Phase record

Phase	Deliverable	Key numbers	Gate
R0	UNIFIED_REGISTRY.csv + schema + corpus citation graph + gap/duplicate reports	2,544 rows; 2,149 papers in citation graph; 1,124 closures live-called with 0 errors; 7 baseline families hash-pinned (LF-normalized)	3146 → 3166
R1	Canonical-route adjudication	109 explicit routes; 2,435 SOLE_ROUTE; 116 Φ -variant tags; 4 author rulings (Φ_{res} sector rule, $\blacksquare$ physical-route standing rule, 2 defect fixes); FRB 0.99c phantom conflict resolved against PAPER_096	3166 → 3173
R2	One corpus pass	199 whitepapers annotated once each (append-only golden rule); pdf2 1,988 → 2,226 PDFs (full parity)	3173 → 3176
R3	uqff_registry_primitives.py single source	24 QCalc *_PRIMITIVE attributes rewired bit-identically; c dual-exposure per §6.2 ruling; 33-entry consumed/vestigial ledger; Python = C++ = Lean pins	3176 → 3186
R4	uqff_registry_graph.py → 656-edge falsifiability graph	SO_5 touches 212 registry rows + 61-site invariant; F_TRZ 111 + 61; D_phys 97; D_crit 43; D_BSFG 41; A_5 23	3186 → 3189
R5	uqff_registry_status.py → status backend + preprint results table + this paper	14 live derived constants, 7 EXACT; residuals: best 0.0000%, median $\approx$ 0.027%, worst 3.08% (H0 — the Hubble tension itself, PAPER_2125)	3189 → (this ship)

All six phases idempotent: every generated artifact carries an identical SHA-256 across repeated runs.

3. The results table (R5 headline)

Computed live from uqff_registry_primitives at build time — the published table cannot drift from the code:

Constant	Canonical route	Closed form	Residual
G	PAPER_593	$(2\pi \cdot D_{\text{crit}}^3 \cdot \Phi_{\text{res}} / (SSq^3 \cdot (26!)^2)) \cdot v_F \blacksquare / (E_0 \cdot f_{\text{THz}})$	0.075%
c	PAPER_592	$(D_{\text{crit}} \cdot 4\pi / \Phi_{\text{res}}) \cdot v_F$	0.10%
μ_0	PAPER_2108	$4\pi \cdot F_{\text{TRZ}} \blacksquare$	EXACT
k_B	PAPER_1209EE S628	$(SSq + \Phi_{\text{5/6}} - F_{\text{TRZ}} \cdot SSq + F_{\text{TRZ}}^2 \cdot D_{\text{phys}} - F_{\text{TRZ}}^2 \cdot SSq) \cdot 10 \blacksquare^{23}$	0.0011%
$\blacksquare$	PAPER_590/1209EE S629	$(D_{\text{BSFG}} + F_{\text{TRZ}} \cdot D_{\text{BSFG}} + F_{\text{TRZ}}^2 \cdot D_{\text{phys}} - F_{\text{TRZ}}^2 \cdot SSq - F_{\text{TRZ}}^2) \cdot 10 \blacksquare^3 \blacksquare / 2\pi$	0.027%
H0	PAPER_2093	$(2 \cdot SO_5 + 2) \cdot F_{\text{TRZ}}^4 \blacksquare$	3.08% = Hubble tension
Λ	PAPER_2094/1156	$(SO_5 + 1) \cdot F_{\text{TRZ}} \blacksquare^3$	0.90%
κ	PAPER_2112	$(SO_5/2) \cdot F_{\text{TRZ}} \blacksquare$	EXACT
B_crit	PAPER_2126	$D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{12}$	EXACT
k_spring	PAPER_1203	$(\rho_{\text{UA}} / \rho_{\text{SCm}}) \cdot \omega_{\text{SCm}} \cdot \Phi_{\text{res}}$	EXACT

λ_{vac}	PAPER_2120	$(\text{SO}_5+1) \cdot \rho_{\text{SCm}}$	EXACT
T_{SCm}	PAPER_1072	$h \cdot f_{\text{SCm}}/k_B$	0.007%
D_{BSFG}	PAPER_1521	$D_{\text{crit}} - 2 \cdot \text{SO}_5$	EXACT
K_{MEX}	PAPER_1522	$\Phi_5/6 \cdot \text{SO}_5/D_{\text{phys}}$	EXACT

All references are observations or SI definitions, reported as observations. Residuals are honest disclosures (Rule 7). The H_0 residual is retained deliberately: the 3.08% gap between the F_{TRZ} -grid value (CMB side) and the local observed anchor is the Hubble tension, addressed by PAPER_1156's 1/12 EXACT tilt closure.

4. Falsifiability, computed rather than claimed

UNIFIED_REGISTRY_FALSIFIABILITY.md (R4) answers the external validator's first question. A revision to SO_5 alone would simultaneously break: 212 registry rows, the 61-site $(\text{SO}_5+1)/\text{SO}_5$ successor-ratio invariant (PAPER_2128), the derived constants F_{TRZ} , D_{BSFG} , K_{MEX} , κ , ρ_{UA} , λ_{vac} , k_{spring} , B_{crit} , and Λ , and the corresponding pins in a 3,180+ assertion gate. Over-determination is the falsifiability mechanism: wrong values cannot hide.

5. Production posture after R5

- **Status reporting:** `uqff_registry_status.py::calculate_status_report()` returns program-level statistics computed live; nothing hand-maintained.
- **Preprint:** UNIFIED_REGISTRY_RESULTS_TABLE.md/.csv is the Tier-1 A3 results table, regenerated from code on demand.
- **API data model:** the registry row schema (quantity, kind, canonical_route, formula, value, reference, residual_pct, sector, phi_variant, paper_source, confirmations, py/cpp/lean sites, corpus_citations, status) is the Tier-3 interchange format.
- **Reproducibility chain:** `registry_generator.py` → `uqff_registry_primitives.py` → `uqff_registry_graph.py` → `uqff_registry_status.py`, each idempotent, each read-only over protected baselines.

6. Cross-references

UNIFIED_REGISTRY_PROGRAM_PLAN.md (the plan this paper closes out); UNIFIED_REGISTRY.csv/_SCHEMA.md/_GRAPH.csv/_FALSIFIABILITY.md/_STATUS_REPORT.md/_RESULTS_TABLE.md; PAPER_592/593 (c, G routes), PAPER_2093/2094/1156 (H_0 , Λ), PAPER_2108 (μ_0), PAPER_2112 (κ), PAPER_1209EE (k_B , ■ compositions), PAPER_590 (■ physical route), PAPER_2126 (B_{crit}), PAPER_1203 (k_{spring}), PAPER_2120 (λ_{vac}), PAPER_1072 (T_{SCm}), PAPER_1521/1522 (D_{BSFG} , K_{MEX} derivatives), PAPER_2128 (61-site invariant), PAPER_2125 (Hubble-tension identification), PAPER_2129 (Φ sector rule).

7. Summary Statement

PAPER_2130 closes the Unified Registry Program. Six phases (R0–R5) converted the Star-Magic corpus into a machine-regenerable replication ledger: 2,544 registered quantities with adjudicated canonical routes, one append-only corpus pass (199 notes, full PDF parity), a single registry-backed primitives module with three-language agreement, a 656-edge falsifiability graph, and a live status/results backend. UQFF's public claim is now infrastructure, not assertion: 9 independent primitives → 14 live-composed derived constants (7 EXACT, honest residuals throughout, the worst residual being the Hubble tension itself) → 2,544 quantities → a 3,180+ assertion gate, reproducible end-to-end by four idempotent scripts over hash-protected baselines.

Filed 2026-07-24. Append-only henceforth.